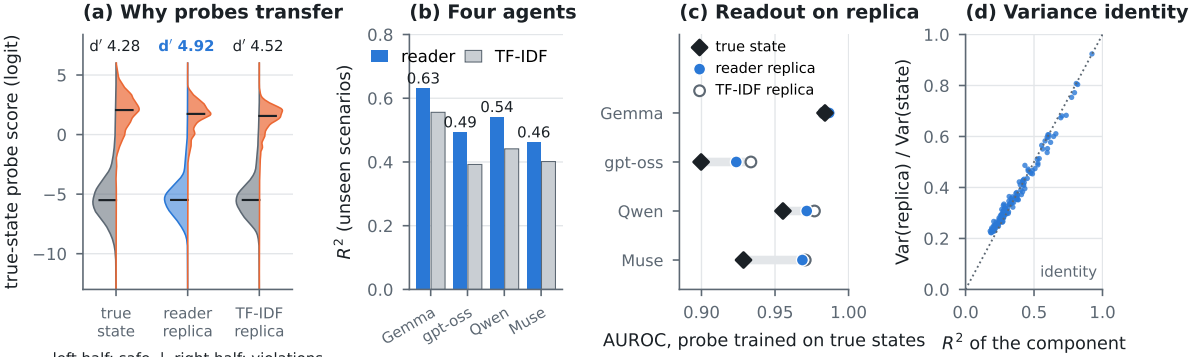

(a) Why probes transfer
d′ 4.28
d′ 4.92
d′ 4.52
true-state probe score (logit)
5
0
−5
−10
true state
reader replica
TF-IDF replica
left half: safe | right half: violations
(b) Four agents
reader
TF-IDF
0.63
0.49
0.54
0.46
R^2 (unseen scenarios)
0.8
0.6
0.4
0.2
0.0
Gemma
gpt-oss
Qwen
Muse
(c) Readout on replica
true state
reader replica
TF-IDF replica
Gemma
gpt-oss
Qwen
Muse
0.90
0.95
1.00
AUROC, probe trained on true states
(d) Variance identity
Var(replica) / Var(state)
1.0
0.8
0.6
0.4
0.2
0.0
identity
0.0
0.5
1.0
R^2 of the component